Blackhole Tracker: Calibrated Compact-Lens Inference and Admissible Boundary Responses in a Conditional BHSM Realization

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Abstract

We present Blackhole Tracker, a runnable research toolkit for mass, center and proper-motion inference from resolved compact-lens image pairs, flux ratios and delays with a full supplied covariance. A local app, command line, Python package and executed notebooks share nonlinear inference, nuisance profiling and explicit rejection or unresolved outcomes. In a fixed synthetic ten-year, three-source validation, all 64 noise realizations complete; nominal local 95% interval coverage is 59–61 of 64 across five tested mass, center and motion components. Short cosmological tracks remain motion unresolved, and an omitted ordinary shear produces a rejected fit. The optional scalar–tensor realization used by the Berger–Hopf Standard Model (BHSM) program supplies a normalized boundary-to-lensing response and a uniform sufficient amplitude bound throughout its retained exterior. It also exposes exact information losses: two projected quadrupole modes vanish, the leading screened primary-pair delay perturbation vanishes, and a standard apparent-mass statistic confounds its leading response with external shear. At an admitted state cap the nearby example's conditional envelope remains far below the supplied noise. The contribution is a validated specialist workflow and its action-specific admission and blind-direction results, not a first compact-lens tracker or empirical black-hole detection. Distances and calibrations are supplied; neither a microscopic BHSM-to-action map nor a horizon flux or physical deformation state is selected.

Keywords: strong gravitational lensing; compact objects; Vainshtein screening; scalar–tensor gravity; boundary response; identifiability

1 The Blackhole Tracker question

Lensing already measures the mass and motion of dark compact objects. Astrometric microlensing has been used to infer an isolated stellar-mass black-hole candidate's properties [1]; joint astrometry and parallax separate familiar mass, distance and proper-motion combinations. Moving dark structures also motivate multi-source astrometric templates [2], and repeated lensed bursts carry motion-dependent delays [3]. The product makes this measurement workflow reproducible under an explicit data contract. Its action-specific question is narrower: can a specified envelope deformation supply a calculable linked pattern in images, fluxes and delays, and which parts survive an ordinary lens fit?

BHSM organizes its research around boundary geometry, constrained actions and envelope matching. That language does not by itself fix a metric, surface state or optical coefficient. An ordinary gravitational potential may already describe the same boundary contribution. We therefore use a single scalar–tensor action and count its metric response once. The name *Blackhole Tracker* denotes the proposed application to a candidate compact dark source. All rays in this

Table 1: Recovered objects relevant to the tracker. "Not located" refers only to the searched record. Exact paths/hashes are in the source map.

Object	Scientific status and use
R1 covariant action and homogeneous background	Conditional effective realization; archived source and hashes recovered and replayed. Supplies the local coefficients used here.
Compact-source cubic reduction	Recovered action-derived quasi-static constraints and regular-body branch. Re-derived and checked here with general exterior scalar flux.
BHSM black-hole/surface drafts	Surface identities and proposed flux parameters recovered. Horizon selection of the R1 exterior flux and causal deformation response not established by those objects.
Microscopic constrained Hessian/Schur work	Actual local response machinery recovered; an evaluated normalized microscopic-to-cosmological optical map was not located.
Fixed external environment	Seven action-owned interface variations remain even when external labels are fixed. This microscopic prescription does not fix a cosmological or black-hole deformation state.

paper lie far outside a declared worldtube, itself far outside the gravitational radius. Lensing on that domain does not establish a horizon or identify the compact interior.

The central deliverable is a usable research product with an ordinary compact-lens tracking mode and an optional conditional boundary-response mode. The new theoretical result is a uniform sufficient admission bound on the normalized quadrupole throughout the retained R1 exterior. The optical calculation identifies exact blind directions and demonstrates why a working compact-mass track need not detect an envelope. A separate nonlinear product validation tests mass/motion recovery, full covariance, limited interval calibration and unsuccessful controls. These statements are independent of accepting a departure from GR; their action-specific application remains conditional.

1.1 Recovered record and scope of the new calculation

A finite recovery covered relevant local Git branches/worktrees, research files in Downloads and Documents, selected members of a mounted BHSM backup, and accessible excerpts of the conversation *Strong Lensing Paper Plan*. The source map and content hashes are distributed with the calculation. This is actual access to selected material, not comprehensive recall of every conversation or archive.

The earlier angular paper was recovered from `BHSM_Angular_Lensing_Reproducibility.zip`, SHA256 beginning `84786b6b905ae1e2`. Its round closed-background, nine-harmonic, six-coordinate calculation is preserved separately. Its illustrative fixed-physical-source SIS delay shift, -0.353216090 s against a 7.753621449 d baseline, is a conditional chosen state, not a selected black-hole signal. The earlier observational HE0435 paper and its failed or incomplete observational tests are also preserved. No such data are used here. A separate Hopf-charge result is retained as supporting research rather than made the central tracker claim.

The strongest feasible tracker route is thus a normalized *conditional effective-action* calculation. We do not substitute a familiar surface mass identity for the missing horizon problem. The spherical flux is kept as an exterior matching datum; the physical state remains free. The mathematical chain, and the place where it stops, are made explicit.

2 Source action, exterior boundary response and optical normalization

This section defines the precise conditional exterior problem used by the tracker. The action, background and source pairing are supplied; a physical black-hole horizon relation and the microscopic BHSM-to-R1 identification are not inferred. The worldtube below is outside the unresolved compact interior and is not a horizon boundary condition.

2.1 Common action and local source

In signature $(-+++)$, initially setting $c = \hbar = 1$ and $M_{\rm Pl}^{-2} = 8\pi G$, the retained action is

$$S = \int \sqrt{-g}\left[\frac{M_{\rm Pl}^2}{2}R - \frac{1}{2}(\nabla T)^2 + \frac{(\nabla T)^2\Box T}{\lambda} - V_0 + \mu^3 T\right] d^4x + S_m. \tag{1}$$

Here physical $\lambda = \Lambda^3$, while the pinned numerical input is $\widehat{\lambda} = \lambda/(M_{\rm Pl}H_0^2) = 1$. This is a cubic kinetic-braiding action in the established scalar–tensor setting [4, 5]. With $\Pi_{\mu\nu} = \nabla_\mu\nabla_\nu T$, variation gives

$$\Box T + \frac{2}{\lambda}\left[\Pi_{\mu\nu}\Pi^{\mu\nu} - (\Box T)^2 + R_{\mu\nu}\nabla^\mu T\nabla^\nu T\right] + \mu^3 = 0. \tag{2}$$

It reproduces the frozen homogeneous current equation $J = v + 6Hv^2/\lambda$, $\dot{J} + 3HJ = \mu^3$, where $v = \dot{\bar{T}}$. Thus the local source sign follows from the same action and background.

For a minimally coupled point-matter model, $S_{\rm pp} = -M\int ds$ and $\delta S_{\rm pp} = \frac{1}{2}\int\sqrt{-g}T_{\rm pp}^{\mu\nu}\delta g_{\mu\nu}d^4x$. In its nonrelativistic local limit the excess density is $M\delta^{(3)}(\boldsymbol{r})$, with no direct scalar insertion $\delta S_{\rm pp}/\delta T$. Strongly self-gravitating horizon response does not follow from treating that point-matter action as a black hole.

Subtract the homogeneous equations, write $T = \bar{T}(t) + \pi_T(r)$ and freeze v, $d = \ddot{\bar{T}}$ and H during a local encounter. At the retained weak-metric, quasi-static order, keeping the nonlinear scalar Hessian terms, the scalar and metric rows are

$$\begin{aligned} A\Delta\pi_T - \frac{2}{\lambda}[(\Delta\pi_T)^2 - |\operatorname{Hess}\pi_T|^2] + b\Delta\Phi = 0, \\ 2M_{\rm Pl}^2\Delta\Phi = \delta\rho - b\Delta\pi_T, \qquad \Phi = \Psi, \\ b = \frac{2v^2}{\lambda}, \qquad A = 1 + \frac{4(d+2Hv)}{\lambda}. \end{aligned} \tag{3}$$

The density is the local excess, not the homogeneous cosmic density counted again. Eliminating that same metric row gives

$$C\Delta\pi_T - \frac{2}{\lambda}[(\Delta\pi_T)^2 - |\operatorname{Hess}\pi_T|^2] = -\frac{b\,\delta\rho}{2M_{\rm Pl}^2}, \qquad C = A - \frac{b^2}{2M_{\rm Pl}^2}. \tag{4}$$

For example, varying the static functional

$$S_{\rm QS} = \int dt\, d^3r\left[-\frac{C}{2}|\nabla\pi_T|^2 + \frac{|\nabla\pi_T|^2\Delta\pi_T}{\lambda} + \frac{b\,\delta\rho\,\pi_T}{2M_{\rm Pl}^2}\right] \tag{5}$$

reproduces Eq. (4). Its last term is the consumed source pairing after metric elimination, not an additional covariant matter coupling. Higher weak-field orders need not retain $\Phi = \Psi$.

2.2 Spherical exterior, scales and open compact-object flux

Let $L_0 = c/H_0$, $x = r/L_0$, $\widehat{m} = (GM/c^2)/L_0$, $\widehat{v} = v/(M_{\rm Pl}H_0)$, and $\widehat{d} = d/(M_{\rm Pl}H_0^2)$. Define

$$\widehat{\beta} = \frac{b}{M_{\rm Pl}} = \frac{2\widehat{v}^2}{\widehat{\lambda}}, \quad C = 1 + \frac{4(\widehat{d} + 2\widehat{H}\widehat{v})}{\widehat{\lambda}} - \frac{\widehat{\beta}^2}{2}, \qquad p(x) = \frac{d(\pi_T/M_{\rm Pl})}{dx}. \tag{6}$$

Regular-origin matching for an ordinary finite matter source gives the exterior flux $-\widehat{\beta}\widehat{m}$. To expose the uncompleted compact-object matching, retain a supplied nonnegative flux ratio χ:

$$\begin{aligned} Cx^2p - \frac{4}{\widehat{\lambda}}xp^2 &= -\chi\widehat{\beta}\widehat{m}, \\ p(x) &= \frac{-2\chi\widehat{\beta}\widehat{m}}{x^2\left[C + \sqrt{C^2 + 16\chi\widehat{\beta}\widehat{m}/(\widehat{\lambda}x^3)}\right]}, \\ \Phi_x = \Psi_x &= \frac{\widehat{m}}{x^2} - \frac{\widehat{\beta}}{2}p(x). \end{aligned} \tag{7}$$

This root is continuous with the decaying linear response. The metric monopole mass remains M; χ changes the scalar flux, not the mass. The value $\chi = 1$ is the regular-matter reference, not a derived black-hole charge. A horizon-regular, cosmologically matched solution is required to determine the physical horizon relation. A negative flux is outside the real, positive-discriminant branch implemented here.

The exterior screening scale and extra force fraction are

$$r_V^3 = \frac{16\chi\widehat{\beta}\,(GM/c^2)L_0^2}{\widehat{\lambda}C^2}, \quad \epsilon_\infty = \frac{\chi\widehat{\beta}^2}{2C}, \quad F(r) = \frac{\delta\Phi'}{(GM/c^2)/r^2} = \frac{2\epsilon_\infty}{1 + \sqrt{1 + (r_V/r)^3}}. \tag{8}$$

The scalar field is not the provider's time-displacement column: $\pi_T = -v\pi_{\rm time}$ when reconstructing it from uniform-field gauge. Adding it directly as another gravitational potential would change both its normalization and the source equation.

At $z_l = 0.3$ the pinned background replay gives

$$\widehat{\beta} = 0.113063844895134, \quad C = 4.23885855943370, \quad D_t = 4.39622102870814, \tag{9}$$

where $D_t = 1 + 12\widehat{H}\widehat{v}/\widehat{\lambda} + 6\widehat{v}^4/\widehat{\lambda}^2$ is the homogeneous leading time principal coefficient. For $M = 10^9 M_\odot$, $H_0 = 67.4$ km/s/Mpc, $\chi = 1$ and $r_0 = 10$ pc, $(L_0, GM/c^2, r_V) = (4.4479593175 \times 10^9, 4.7854158416 \times 10^{-5}, 45680.3117591)$ pc. The compactness at the worldtube is 4.7854×10^{-6} and $H_l r_V/c = 1.2156 \times 10^{-5}$. These are local expansion parameters, not measured error bars. Near a horizon the weak-metric approximation fails; the worldtube must remain outside that domain.

2.3 Exact linear multipoles and canonical boundary response

Write the dimensionless scalar perturbation as $\delta\widehat{\pi} = af_\ell(r)Y_{\ell m}$, with $\int |Y_{\ell m}|^2 d\Omega = 1$. Linearizing Eq. (4) gives

$$\delta E = Z^{ij}\partial_i\partial_j\delta\pi = \partial_i(Z^{ij}\partial_j\delta\pi), \qquad Z^{ij} = C\delta^{ij} - \frac{4}{\lambda}[(\Delta\pi_0)\delta^{ij} - \Pi_0^{ij}], \tag{10}$$

because $\partial_i Z^{ij} = 0$. The anisotropic Hessian term is essential. The spherical eigenvalues satisfy

$$K_r = C - \frac{8\pi_0'}{\lambda r}, \quad K_\Omega = C - \frac{4}{\lambda}(\pi_0'' + \pi_0'/r), \quad K_r' = \frac{2(K_\Omega - K_r)}{r}. \tag{11}$$

Outside the source the normalized radial equation is therefore

$$\frac{1}{r^2}(r^2 K_r f_\ell')' - \frac{\ell(\ell+1)K_\Omega}{r^2} f_\ell = 0. \tag{12}$$

An actual density multipole adds the fixed source $-b\delta\rho_{\ell m}/(2M_{\rm Pl}^2)$ to the right side and requires interior matching. A prescribed boundary value is not that interior solve.

For $\rho = r/r_V$ and $u = \rho^{-3}$, the exact exterior coefficients are

$$K_r = C\sqrt{1+u}, \quad K_\Omega = C\frac{1+u/4}{\sqrt{1+u}}, \quad K_t = D_t + \frac{3(S-C)^2}{4S} \geq D_t > 0, \quad S = C\sqrt{1+u}. \tag{13}$$

Thus this retained scalar problem is elliptic and has a positive leading time coefficient. This is not full-system stability or a horizon/UV certificate; its scalar characteristic speeds can exceed the metric light speed in part of the exterior.

The decaying solution of Eq. (12) is

$$R_\ell(\rho) = \rho^{-\ell-1}{}_2F_1\left(\frac{\ell+1}{2}, \frac{\ell+2}{6}; \frac{2\ell+4}{3}; -\rho^{-3}\right), \qquad f_\ell(r) = \frac{R_\ell(r/r_V)}{R_\ell(r_0/r_V)}. \tag{14}$$

It follows by substituting $t = -\rho^{-3}$ in the radial equation. The deep-screened powers are $(\ell+1)/2$ and $-\ell/2$; the far powers are ℓ and $-\ell-1$. In particular the decaying quadrupole has r^{-1} and r^{-3} limits. For $\ell = 1$, ${}_2F_1(1, 1/2; 2; -u) = 2/(1+\sqrt{1+u})$, so f_1 is exactly proportional to $-p$: translating the monopole reproduces the linear dipole equation. This supplies an independent field-equation check. The $\chi = 0$ limit is $f_\ell = (r_0/r)^{\ell+1}$.

The exterior quadratic energy, divided by its overall $M_{\rm Pl}^2 L_0$ factor, is

$$\frac{1}{2}\int_{x_0}^{\infty} [x^2 K_r(\partial_x \delta\widehat{\pi})^2 + \ell(\ell+1)K_\Omega \delta\widehat{\pi}^2]dx = \frac{1}{2}\Lambda_\ell a^2, \qquad \Lambda_\ell = -x_0^2 K_r(x_0) f_{\ell,x}(x_0) > 0. \tag{15}$$

The inner normal of the exterior domain is $-\partial_r$; its conjugate flux is $q_{\ell m} = \Lambda_\ell a$. Positivity establishes uniqueness of the zero-data decaying exterior solution. Holding a fixed and holding $q_{\ell m}$ fixed are different source comparisons. Neither amplitude is selected by the action without an interior/boundary state. The numerical reference has $\Lambda_2 = 0.00294226449484$ in these dimensionless-x conventions.

The same metric constraint yields $\delta\Phi = \delta\Psi = \delta\Phi_N - \widehat{\beta}\delta\widehat{\pi}/2 + H_{\rm match}$, where $\Delta H_{\rm match} = 0$. Ordinary gravitational multipoles and outer matching remain independent until a common source fixes them. The implemented scalar-driven term adds no arbitrary metric quadrupole.

2.4 Projected phase, derivatives and finite matching

For a transverse real quadrupole choose

$$Y_N = \sqrt{\frac{15}{16\pi}}(n_x^2 - n_y^2) \equiv N_2(n_x^2 - n_y^2), \qquad W \equiv \Phi + \Psi = -\widehat{\beta} a f_2(r) Y_N. \tag{16}$$

For impact $b \geq r_0$, every integration point lies outside the worldtube. The scalar projected physical phase length is

$$\begin{aligned} \delta\widehat{\psi}(b,\varphi) &= a\, q(b)\cos 2\varphi, \\ q(b) &= -\widehat{\beta} N_2 \int_{-\infty}^{\infty} f_2(\sqrt{b^2+z^2})\frac{b^2}{b^2+z^2}\, dz \\ &= -2\widehat{\beta} N_2 b \int_0^{\pi/2} f_2(b\sec\theta) d\theta. \end{aligned} \tag{17}$$

Both halves of the line are included. With $r = b\sec\theta$,

$$q'(b) = -2\widehat{\beta}N_2\int_0^{\pi/2}(f_2 + rf_2')d\theta,$$
$$q''(b) = -\frac{2\widehat{\beta}N_2}{b}\int_0^{\pi/2}(r^2f_2'' + 2rf_2')d\theta. \tag{18}$$

The provider returns q, q', q'' in pc, dimensionless units and pc^{-1}, respectively. Exact hypergeometric connection formulae preserve the cancellation of the deep r^{-1} contribution in these radial derivatives. Projection gives a deep phase plateau and a far b^{-2} phase, rather than identifying a local force ratio with a projected photon deflection.

An optional axial boundary coordinate uses the independently normalized $Y_{20} = N_0(3n_z^2 - 1)$, $N_0 = \sqrt{5/(16\pi)}$. Its physical phase per unit axial boundary amplitude is

$$q_0(b) = -2\widehat{\beta}N_0 b\int_0^{\pi/2}(2\sec^2\theta - 3)f_2(b\sec\theta)d\theta. \tag{19}$$

Its first and second radial derivatives replace f_2 by $f_2 + rf_2'$ and $r^2f_2'' + 2rf_2'$, respectively, with prefactors $-2\widehat{\beta}N_0$ and $-2\widehat{\beta}N_0/b$. This supplies a separately normalized radial nuisance when the axial state is not fixed. It is distinct from varying the spherical scalar flux. Its leading deep-screened phase is logarithmic and can be mass-like. In the exact zero-flux harmonic limit its full-line projection vanishes; the transverse phase is then exactly $q = -4\widehat{\beta}N_2r_0^3/(3b^2)$. The implementation uses these analytic zero-flux identities instead of an interpolation residual.

The scalar monopole deflection is supplied separately:

$$\delta\widehat{\alpha}_0(b) = -\frac{\widehat{\beta}b}{L_0}\int_{-\infty}^{\infty}\frac{p(r/L_0)}{r}\,dz = \frac{4GM}{c^2b}\int_0^{\pi/2}F(b\sec\theta)\cos\theta\,d\theta. \tag{20}$$

The ordinary $4GM/(c^2b)$ is counted once by the lens model. A relative spherical phase is obtained by integrating this deflection between impacts; its formally divergent absolute line potential is unnecessary.

Infinity in these expressions is a matched thin-lens extension, not an exact static spacetime. A local overlap cutoff obeys $\max(b, r_V) \ll L \ll c/H_l, R_{\mathrm{curvature}}$. The scalar monopole tail, divided by $4GM/(c^2b)$, is bounded at leading order by $\epsilon_\infty b^2/(2L^2)$. If $A_2 = r_V^3/R_2(r_0/r_V)$, the Euler integral of the hypergeometric function gives $f_2(r) \le A_2/r^3$ and hence

$$|q_{|z|>L}| \le \frac{\widehat{\beta}N_2A_2b^2}{2L^4}. \tag{21}$$

These control the stated tail only. They do not bound coefficient time dependence, a changed environment, or a different physical outer state.

2.5 Perturbative example and actual numerical control

For the supplied state convention $a = \eta x_0|p(x_0)|/6$, $\eta = 10^{-3}$, the numerical reference has $a = 1.37809214552 \times 10^{-16}$. The closed R1 distances at $(z_l, z_s) = (0.3, 1)$ give an Einstein impact 339.621287478 pc and $aq(r_E)/(4GM/c^2) = -8.891115935 \times 10^{-13}$. These amplitudes are declared inputs, not horizon-selected values.

On 65 radii from r_0 to $10r_V$ and 100 directions, the sampled maximum perturbation/background scalar-Hessian norm ratio is 3.742×10^{-4}. The exact linearized equation residual, scaled by its principal Hessian terms, is below 6.1×10^{-15}. The positive comparison scale used here is

$$\mathcal{S} = \left[C + \frac{4}{\lambda}(|\operatorname{tr}\Pi_0| + \|\Pi_0\|_F)\right]\|\delta\Pi\|_F. \tag{22}$$

The retained nonlinear scalar remainder is exactly $-2[(\operatorname{tr}\delta\Pi)^2 - \delta\Pi : \delta\Pi]/\lambda$; its sampled ratio to the stated principal-term scale is at most 8.57×10^{-5}. These are finite-sample diagnostics, not uniform interval certificates.

Independent radial integration agrees with the special function to 2.25×10^{-10} relative over the tested domain, with the outer decaying condition imposed at $1000 r_V$. The exact translation check is within 1.23×10^{-15}. Independent original-line quadrature checks the projected quadrupole phase to 2.23×10^{-16} pc and the scalar monopole deflection to 4.04×10^{-27} at $b = 1000$ pc. The code records Gauss-order refinement and Hermite interpolation discrepancies for every prepared table. Its fast potential, gradient and Hessian derive from one interpolating potential, preserving their mutual consistency.

Slow source motion requires small velocity relative to the scalar characteristic speeds; a varying multipole additionally requires a quasi-static crossing condition such as $\omega r_V / c_{\mathrm{scalar}} \ll 1$. The retained field equations omit higher weak-metric and time-dependent terms. A nonzero environmental scalar Hessian modifies the same operator and cannot generally be added by linear superposition. The paper's homogeneous-background exterior avoids that extra assumption. The positive principal coefficients, small illustrative amplitude and tiny formal delay do not establish physical precision at that delay, an observational forecast, or a solved black-hole boundary problem.

3 A uniform admission bound for boundary states

The exterior action gives a sufficient amplitude domain before any optical fit. Let $\mathcal{H}_0 = \mathrm{Hess}_x \widehat{\pi}_0$ and $\delta\mathcal{H} = \mathrm{Hess}_x \delta\widehat{\pi}$, with derivatives in $x = r/L_0$. In an orthonormal real quadrupole basis, write

$$\delta\widehat{\pi} = f_2 \sum_{A=1}^{5} a_A Y_A = N_{\mathrm{STF}} f_2 \, n^i \mathsf{Q}_{ij} n^j, \qquad N_{\mathrm{STF}} = \sqrt{\frac{15}{8\pi}}, \quad \mathcal{A} = \|\boldsymbol{a}\|_2 = \|\mathsf{Q}\|_F, \tag{23}$$

where Q is symmetric and trace free and $f_2(x_0) = 1$. The result covers all five coordinates, including the two erased by a symmetric line-of-sight projection. A smaller template family specifies its omitted coordinates separately.

Proposition 1 (Uniform retained-exterior control)**.** *Assume the isolated decaying branch, $\chi > 0$, $P_0 > 0$, and $C, D_t, \widehat{\lambda} > 0$. Put $P_0 = x_0|p(x_0)|$, $B = 12 N_{\mathrm{STF}}$, $d_* = D_t/C$, and*

$$R_t = \begin{cases} 6, & d_* \geq 3/4, \\ 6t(2-t), \quad t = \sqrt{4d_*/3}, & 0 < d_* < 3/4. \end{cases} \tag{24}$$

At every $x \geq x_0$ and every angular direction,

$$\frac{\|\delta\mathcal{H}\|_F}{\|\mathcal{H}_0\|_F} \leq \frac{B\mathcal{A}}{\sqrt{2}P_0}, \qquad \frac{\|\delta Z\|_{\mathrm{op}}}{K_\Omega} \leq \frac{2\sqrt{2}B\mathcal{A}}{P_0}, \qquad \frac{|\delta K_t|}{K_t} \leq \frac{4\sqrt{3}B\mathcal{A}}{R_t P_0}. \tag{25}$$

Here $\delta Z = -4[(\operatorname{tr}\delta\mathcal{H})I - \delta\mathcal{H}]/\widehat{\lambda}$ and $\delta K_t = -4 \operatorname{tr}\delta\mathcal{H}/\widehat{\lambda}$ are the retained principal perturbations.

Proof. Set $k = K_\Omega/K_r \in [1/4, 1]$ and $y = d\log f_2/d\log r$. Equation (12) gives

$$r^2 f_2'' + 2kr f_2' - 6k f_2 = 0, \qquad \frac{dy}{d\log r} = (1-2k)y + 6k - y^2. \tag{26}$$

At $y = -3$ and -1 the outward vector fields are respectively $12(k-1) \leq 0$ and $8k - 2 \geq 0$. Thus $[-3, -1]$ is invariant under inward integration and contains the decaying far-field branch

$y = -3 + 9(r_V/r)^3/8 + O((r_V/r)^6)$. Hence $-3 \le y \le -1$. The elliptic maximum principle gives positive f_2 and also

$$f_2(x) \le f_1(x) = \frac{|p(x)|}{|p(x_0)|}. \tag{27}$$

Indeed $L_2 f_1 = -4K_\Omega f_1/r^2 < 0$, whereas $L_2 f_2 = 0$; their normalized inner and decaying outer boundary data agree.

Rotate the evaluation direction to the third axis. The Hessian map on the orthogonal $m = 2, 1, 0$ matrix subspaces has singular coefficients

$$2, \qquad 2|y-1|, \qquad \frac{2}{\sqrt{3}}\sqrt{1+2k^2}\,(3-y), \tag{28}$$

after removing $N_{\mathrm{STF}} f_2/x^2$. Each is at most 12, so $\|\delta\mathcal{H}\|_F \le B\mathcal{A} f_2/x^2$. Now use Eq. (27), $\|\mathcal{H}_0\|_F \ge \sqrt{2}|p|/x$, $\|(\operatorname{tr} H)I - H\|_{\mathrm{op}} \le \sqrt{2}\|H\|_F$, and $K_\Omega \ge 2|p|/(\widehat{\lambda}x)$. Finally, with $s = K_r/C \ge 1$, Eq. (13) gives

$$\frac{K_t}{|p|/(\widehat{\lambda}x)} = 6 - \frac{6}{s} + \frac{8d_*}{s-1} \ge R_t. \tag{29}$$

Minimization over s yields Eq. (24); $|\operatorname{tr} H| \le \sqrt{3}\|H\|_F$ completes the result. □

For declared budgets $\epsilon_H > 0$, $0 < \epsilon_Z < 1$ and $\eta_{\max} > 0$, a sufficient state cap is

$$\mathcal{A} \le \mathcal{A}_{\max} = P_0 \min\left\{\frac{\eta_{\max}}{6}, \frac{\sqrt{2}\epsilon_H}{B}, \frac{\epsilon_Z}{2\sqrt{2}B}, \frac{\epsilon_Z R_t}{4\sqrt{3}B}\right\}, \qquad \eta = \frac{6\mathcal{A}}{P_0}. \tag{30}$$

This preserves the retained principal signs throughout the exterior and bounds the relative Hessian. These are approximation budgets, rather than a selected astrophysical amplitude. The exact quadratic scalar remainder obeys $|\mathcal{R}_2| \le 4\|\delta\mathcal{H}\|_F^2/\widehat{\lambda}$; a residual bound alone is not an inverse-operator error estimate. The conormal map is exact for the linear exterior:

$$\boldsymbol{q} = \Lambda_2 \boldsymbol{a}, \qquad \boldsymbol{a} = \boldsymbol{q}/\Lambda_2, \qquad \frac{E_2}{M_{\mathrm{Pl}}^2 L_0} = \frac{1}{2}\Lambda_2 \mathcal{A}^2. \tag{31}$$

Thus specified conormal data are admitted when $\|\boldsymbol{q}\|_2 \le \Lambda_2 \mathcal{A}_{\max}$. This source pairing does not select a horizon flux or interior state.

At $\chi = 0$, the relative normalization P_0 vanishes but nonzero boundary states remain possible. Here $f_2 = (x_0/x)^3$, $K_r = K_\Omega = C$, $K_t = D_t$, and $\Lambda_2 = 3Cx_0$. The absolute cap

$$\mathcal{A} \le \epsilon_Z \widehat{\lambda} x_0^2 \min\left\{\frac{C}{4\sqrt{2}B}, \frac{D_t}{4\sqrt{3}B}\right\} \tag{32}$$

preserves the same retained principal signs; relative η is undefined.

Admission also requires weak metric potentials, rays with $b \ge r_0$, and a local patch whose background changes slowly. A useful uniform spatial-gradient control is $[(|p_0| + \sqrt{6}N_{\mathrm{STF}}\mathcal{A}/x_0)/|\widehat{v}|]^2 \ll 1$. For a patch scale L, the implemented drift estimate is $\max(|\widehat{H}|, 2|\widehat{d}/\widehat{v}|)L/L_0$. The retained sound-speed lower bound follows by minimizing $\min(4s^2, s^2+3)/[3(s-1)^2 + 4d_* s]$ over $s \ge 1$. Writing its square root as $c_{\min}$ in units of c, require $v_\perp/(cc_{\min}) \ll 1$ and $L/(cc_{\min}\tau_{\mathrm{local}}) \ll 1$ for intrinsic evolution. Observer proper motion gives $v_\perp = |\mu_{\mathrm{obs}}| D_l (1 + z_l)$; the observer timescale is $\tau_{\mathrm{obs}} = (1+z_l)\tau_{\mathrm{local}}$. Fixed boundary coefficients are an explicit time assumption. The product uses IAU nominal $GM_\odot$ for its mass-length conversion; this unit convention leaves the equations and bounds unchanged. These controls apply to the retained local equations. Cosmological matching, the coupled horizon problem and ultraviolet completion require their own calculations.

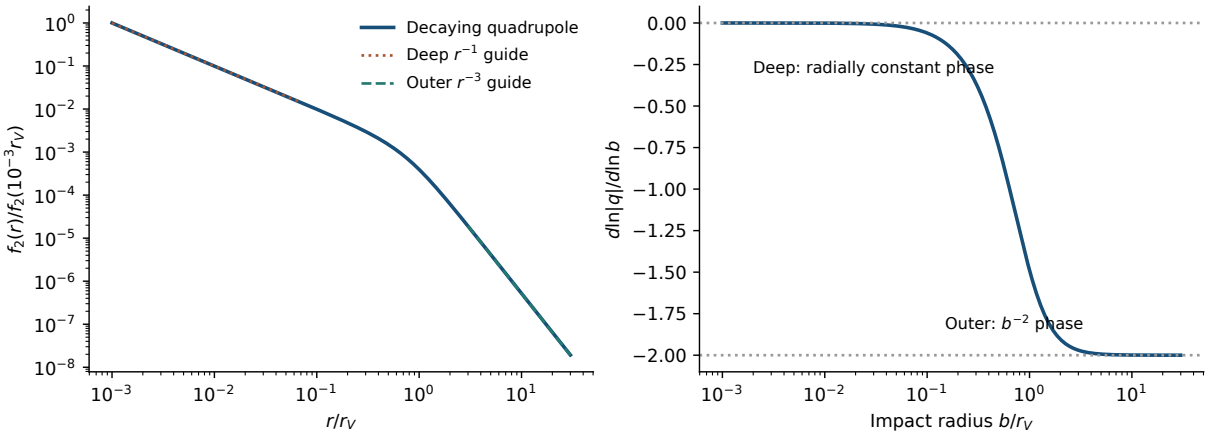


Figure 1: Conditional exterior transfer for the supplied regular-body flux proxy. Left: normalized decaying quadrupole, with its known deep r^{-1} and outer r^{-3} powers. Right: the slope of the fully matched projected phase, approaching zero and -2. The curves use the action-fixed coefficients and the full line integral, not an extrapolation of either power law across the transition.

4 Projection and the information retained by the tracker

The screened multipole exponents and hypergeometric radial modes have cubic-Galileon precedents [6–8]. Here the input is the normalized exterior response of the preceding section, with its fixed scalar-to-metric coefficient. The statements below concern that conditional response, rather than a horizon-selected amplitude.

4.1 A normalized STF boundary state and its projected nulls

Let C be a real symmetric trace-free 3×3 matrix, normalized by $\operatorname{tr}\mathsf{C}^2 = 1$, and define

$$Y_{\mathsf{C}}(\mathbf{n}) = N_{\rm STF} C_{ij} n_i n_j, \qquad N_{\rm STF} = \sqrt{\frac{15}{8\pi}}, \qquad \int_{S^2} Y_{\mathsf{C}}^2 \, d\Omega = 1. \tag{33}$$

Thus $a f_2(r) Y_{\mathsf{C}}$ has boundary L^2 norm $|a|$ at r_0. For $\mathsf{C}_+ = \operatorname{diag}(1,-1,0)/\sqrt{2}$, this is precisely the transverse harmonic with $N_2 = \sqrt{15/(16\pi)}$ used above; $N_{\rm STF}/\sqrt{2} = N_2$. Define $\mathcal{W} = \Phi + \Psi$. Its scalar-driven response is $\delta\mathcal{W} = \mathcal{A}_W f_2 Y_{\mathsf{C}}$, with $\mathcal{A}_W = -\widehat{\beta} a$ for $\pi_T/M_{\rm Pl}$ normalization. This is the fixed action coefficient, not an additional photon coupling.

For a frozen straight ray with $\mathbf{x} = (b\cos\varphi, b\sin\varphi, z)$, $b \geq r_0$, factor out $N_{\rm STF}\mathcal{A}_W$ and the known geometric prefactor. The remaining full-line projection is

$$\mathcal{P}_f[\mathsf{C}](b,\varphi) = \int_{-\infty}^{\infty} f(r) \frac{C_{ij} x_i x_j}{r^2} \, dz, \qquad r^2 = b^2 + z^2. \tag{34}$$

Parameterize the five STF components by

$$\mathsf{C} = \begin{pmatrix} -c/2 + s_1 & s_2 & u \\ s_2 & -c/2 - s_1 & v \\ u & v & c \end{pmatrix}, \qquad \operatorname{tr}\mathsf{C}^2 = \frac{3}{2}c^2 + 2(s_1^2 + s_2^2 + u^2 + v^2). \tag{35}$$

Direct integration gives

$$\mathcal{P}_f[\mathsf{C}] = c A_0(b) + T(b)(s_1 \cos 2\varphi + s_2 \sin 2\varphi), \tag{36}$$

$$T(b) = b^2 \int_{-\infty}^{\infty} \frac{f(r)}{r^2} \, dz, \qquad A_0(b) = \int_{-\infty}^{\infty} f(r) \left(1 - \frac{3b^2}{2r^2}\right) dz. \tag{37}$$

The xz and yz terms are odd in z and vanish. This exact two-dimensional kernel prevents reconstruction of an unrestricted three-dimensional STF state from this one-view projection. If its three visible coefficients are inferred with the stated action normalization, they only imply

$$a^2 \geq \frac{3}{2}(ac)^2 + 2[(as_1)^2 + (as_2)^2]. \tag{38}$$

Conditional rank-three result. For the isolated cubic response with

$$K_r = C_0\sqrt{1+(r_V/r)^3}, \quad K_\Omega = K_r + \frac{r}{2}K_r', \quad C_0 > 0, \quad r_V > 0, \tag{39}$$

the normalized positive decaying solution obeys $T(b) > 0$ and $A_0(b) > 0$. Hence (36) has rank three as a map into projected functions, with kernel exactly (u, v). To prove the axial assertion, put $Y = r^3 f$. The radial equation implies

$$(K_r r^{-4} Y')' = 6K_r' r^{-5} Y. \tag{40}$$

Positivity of f follows from the elliptic maximum principle. Since $K_r' < 0$ and $K_r r^{-4} Y' \to 0$ at infinity, integration from r to infinity gives $Y' > 0$. An integration by parts, using $zf(r) \to 0$, then gives

$$A_0(b) = \frac{1}{2}\int_{-\infty}^{\infty} \frac{z^2}{r^2}[rf'(r) + 3f(r)]\,dz = \frac{1}{2}\int_{-\infty}^{\infty} \frac{z^2}{r^4} Y'(r)\,dz > 0. \tag{41}$$

The genuinely screened hypothesis matters: for $f = C_o/r^3$, $A_0 \equiv 0$ and $T = 4C_o/(3b^2)$, so the rank is two. This harmonic limit agrees with the transverse STF mass-multipole projection [9]. Rank in a projected representation does not by itself establish rank in lens data after monopole and source freedoms are fitted.

These full-line statements use the isolated frozen exterior as a local matching model. Finite observer/source endpoints, nonconstant geometric weights, motion during ray crossing, and cosmological outer completion require their own projection. They cannot be assumed to remove the nulls without calculation.

4.2 Radial discrimination and its limitation

In a screened overlap $r_0 \leq b \ll r_V$, the matched $f \sim A_d/r$ gives

$$T(b) \sim 2A_d. \tag{42}$$

The leading transverse potential is therefore constant in radius and spin two in angle. The deep axial integral, temporarily cut off at a symmetric Z, is

$$A_0(b; Z) = A_d\left[2\,\mathrm{asinh}(Z/b) - \frac{3Z}{\sqrt{Z^2+b^2}}\right] \sim A_d[2\ln(2Z/b) - 3]. \tag{43}$$

Outer matching fixes the constant and subleading corrections. The leading $-2A_d \ln b$ is degenerate with a point-mass potential. Its additive constant has no deflection or differential-delay signal. The numerical calculation uses the matched profile rather than extending r^{-1} to infinity.

At the potential level, a constant spin-two coefficient differs from an isolated compact harmonic quadrupole proportional to b^{-2} and a uniform external shear proportional to b^2. If a calibrated spin-two potential amplitude is restricted to

$$p_2(b) = q + \mu b^{-2} + \gamma b^2, \qquad D = b\partial_b, \tag{44}$$

then

$$(D^2 - 4)p_2 = -4q. \tag{45}$$

At three distinct positive radii, writing $x_i = b_i^2$, the design rows $(1, x_i^{-1}, x_i)$ have determinant

$$-\frac{(x_2 - x_1)(x_3 - x_1)(x_3 - x_2)}{x_1 x_2 x_3} \neq 0. \tag{46}$$

Two radii cannot separate q from both nuisance coefficients. For the actual transfer the first column is $T(b_i)$, and its determinant and conditioning must be evaluated. These are conditional potential-reconstruction statements. They are not a claim that three image radii directly measure three potential values. Internal/external multipole reconstruction already has an extensive lensing literature [10, 11].

An arbitrary extended matter distribution is a stronger nuisance family: $\nabla_\perp^2[q\cos 2\varphi] = -4q\cos 2\varphi/b^2$, so an annular distributed quadrupolar density can reproduce this potential. A small quadrupolar perturbation can be supported by a positive monopole. Consequently the radial filter does not identify a gravity theory or exclude an ordinary extended envelope.

4.3 Timing blindness, scalar-estimator degeneracy, and image geometry

Use dimensionless reference potential $\psi_0 = \ln r$, source $\mathbf{y} = u\mathbf{e}$, $s = \sqrt{u^2+4}$ and primary images $\mathbf{x}_+ = a\mathbf{e}$, $\mathbf{x}_- = -d\mathbf{e}$, where $a = (u+s)/2$, $d = a^{-1}$. For a perturbation $q\cos 2(\varphi - \varphi_q)$ constant in radius, both unperturbed image potentials agree. Fermat stationarity therefore gives zero first-order delay change at fixed source. Its log flux-ratio change is

$$\delta \ln R = -4q\cos 2(\varphi_s - \varphi_q)\frac{s}{u}. \tag{47}$$

Tangential image displacements can retain the angular component absent from this scalar correction. These statements concern continued nonsingular primary branches, with perturbations small enough to avoid a critical crossing or flux-order reversal.

The first-variation operator of the point-lens apparent-mass estimator satisfies

$$\mathcal{K}_1(u) = \mathcal{K}_{r^2}(u) = \frac{s^3}{uh(u)}, \qquad h(u) = \frac{1}{2}us + 2\,\mathrm{asinh}(u/2). \tag{48}$$

Thus $(r^2-1)\cos 2\varphi$ is an exact first-order null of this scalar statistic for every $u > 0$. A deep constant spin-two response and external shear cannot be separated by that statistic alone. The point-lens delay/flux relation is established prior art [12]; the evaluated transfer response, rather than different radial powers by themselves, determines its information.

Fixed-data gradient proposition. Let observed image positions $\boldsymbol{\theta}_\pm$ and reference Jacobians $A_\pm$ be fixed, and hold the time-delay prefactor fixed. A perturbing potential p can preserve these positions with one freely shifted source if and only if

$$\nabla p(\boldsymbol{\theta}_+) = \nabla p(\boldsymbol{\theta}_-) = -\delta\boldsymbol{\beta}. \tag{49}$$

Under (49), the remaining first-order null tests are

$$\delta\Delta\tau = (\boldsymbol{\theta}_+ - \boldsymbol{\theta}_-)\cdot\delta\boldsymbol{\beta} + p_+ - p_-, \tag{50}$$

$$\delta \ln R = \mathrm{tr}(A_+^{-1}Hp_+) - \mathrm{tr}(A_-^{-1}Hp_-). \tag{51}$$

The lens equations and Fermat stationarity prove these identities directly. In a forward calculation at fixed source, the images move and the flux derivative also includes the displaced background Hessian; that term cannot be dropped.

For the scalar-estimator null $p = (r^2-1)\cos 2\varphi$, direct differentiation gives

$$\nabla p(a\mathbf{e}) - \nabla p(-d\mathbf{e}) = 2s\cos 2\varphi_s\,\mathbf{e}. \tag{52}$$

With a calibrated center, a free source alone cannot absorb this difference when $\cos 2\varphi_s \neq 0$. An added mass perturbation $\delta m \ln r$ cancels it only for $\delta m = -2\cos 2\varphi_s$, which then changes the mass estimator by δm. Thus image geometry and the estimator distinguish this particular difference if mass is the only additional compensator and the time-distance conversion is fixed. Equivalently, the geometric angular-mass product of the two radii changes by zero for constant spin two and by $2\cos 2\varphi_s$ for r^2 spin two. Comparing angular and physical masses requires distance calibration.

For $\cos 2\varphi_s = 0$, the same difference is a genuine first-order joint null for a single freely positioned source: its gradients agree, its source shift is tangential, and both delay and flux-ratio variations vanish. Additional epochs with changing angle can change this rank. Moving centers, convergence, distance factors, ordinary multipoles, and calibration parameters must be included before profiling the full data Jacobian. More general source-position and multi-plane transformations impose further limits [13–16]. No unrestricted reconstruction theorem follows from the potential-space filter.

4.4 Incoming, local and outgoing optical transfer

We use the homogeneous closed R1 background for the incoming and outgoing distances. If $K = 0.018$ in units $L_0 = c/H_0$, define

$$\chi(z) = \int_0^z \frac{dz'}{\widehat{H}(z')}, \qquad S_K(\chi) = \frac{\sin(\sqrt{K}\chi)}{\sqrt{K}}, \quad D_{ab} = \frac{L_0 S_K(\chi_b - \chi_a)}{1+z_b}. \tag{53}$$

The local deflection then enters the usual single-plane composition,

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \frac{D_{ls}}{D_s}\widehat{\boldsymbol{\alpha}}(D_l(\boldsymbol{\theta} - \boldsymbol{c})), \qquad \widehat{\boldsymbol{\alpha}} = \nabla_b \int (\Phi + \Psi)\, dz. \tag{54}$$

This is the homogeneous limit of general optical transfer, whose equivalent-lens and source-transformation freedoms must be retained [14–16]. No angular environmental harmonic from the earlier cosmological paper is added here: a second nonlinear scalar Hessian background would change the local response operator and require a new reduction.

For a reference mass M, let $m = GM/c^2$,

$$r_E^2 = 4m\frac{D_l D_{ls}}{D_s}, \qquad \boldsymbol{x} = \frac{D_l(\boldsymbol{\theta} - \boldsymbol{c})}{r_E}, \qquad t_M = \frac{4m}{c}(1+z_l). \tag{55}$$

The dimensionless lens potential is the projected Weyl phase divided by $4m$. In particular, the ordinary monopole is $\ln r$, $r = |x|$; the scalar monopole correction is derived from the same action. It is added to that GR metric once. If $q(b)$ is the phase in physical length per unit transverse boundary amplitude,

$$\delta\psi = \epsilon_+ Q(r)\cos 2\varphi + \epsilon_\times Q(r)\sin 2\varphi, \quad Q(r) = \frac{q(r_E r)}{q(r_E)}, \quad \epsilon_A = \frac{a_A q(r_E)}{4m}. \tag{56}$$

The a_A are supplied scalar states, not extra optical couplings. Homogeneous metric quadrupoles, which need not be fixed by a_A, are retained as ordinary compact-lens nuisance parameters.

5 A tracker statistic and its blind direction

For a point source lensed by an isolated point mass, let $R > 1$ be the absolute primary-image flux ratio. The familiar delay law [12] gives

$$u = 2\sinh(\ln R/4), \qquad s = \sqrt{u^2+4},$$
$$h(u) = \tfrac{1}{2}us + 2\,\mathrm{asinh}(u/2), \qquad M_{\mathrm{app}} = \frac{c^3\Delta t}{4G(1+z_l)h(u)}. \tag{57}$$

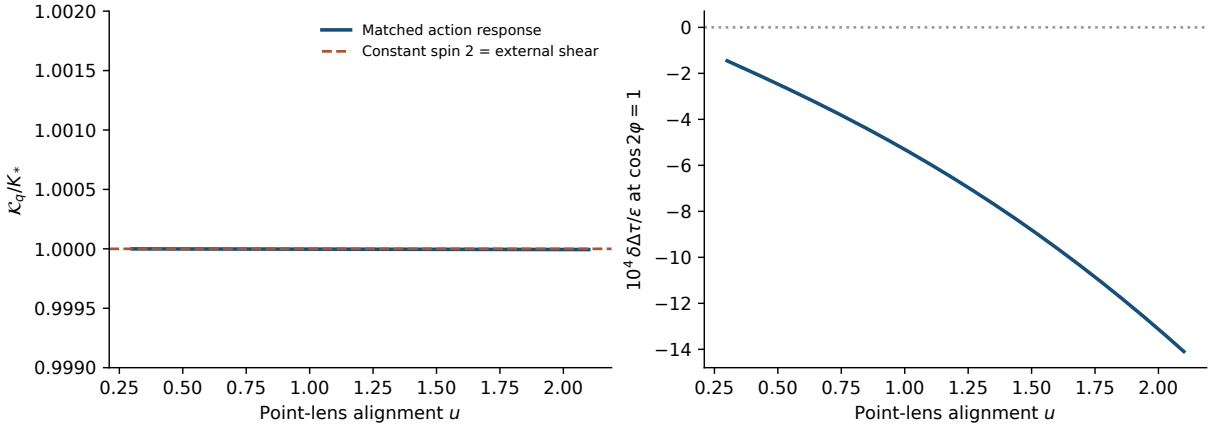


Figure 2: The matched radial profile in the point-mass first-variation operator, with $r_E/r_V = 0.00743$. Left: its apparent-mass response is close to the exactly coincident constant-phase/external-shear response, despite different potentials. Right: the small matching-induced delay derivative; the strict constant-phase result is zero. Curves are derivatives per dimensionless phase amplitude, not physical observed signals.

The apparent mass is constant for any slow constant-mass point-lens trajectory, including source or observer motion incorporated in u. This baseline estimator is inherited. It also remains equal to the point mass in a constant isotropic convergence in the ideal point-lens model: rescaling the image/source coordinates leaves the dimensionless point-mass delay difference and flux relation intact. It is not a general observable invariant for arbitrary lenses.

For a small additional potential ϵp, Appendix A derives

$$\delta \ln M_{\rm app} = \frac{\delta\Delta\tau}{h(u)} - \frac{s^2}{4h(u)}\delta \ln R. \tag{58}$$

Image displacements must enter the magnification derivative even though Fermat stationarity removes them from the first-order delay at fixed source. For $p = q(r)\cos 2\varphi$, write its response as $\delta \ln M_{\rm app} = \mathcal{K}_q(u)\cos 2\varphi_s$. An exact first-order identity is

$$\mathcal{K}_1 = \mathcal{K}_{r^2} = K_*(u), \qquad \mathcal{K}_{r^{-2}} = (u^2+1)K_*(u), \qquad K_* = \frac{(u^2+4)^{3/2}}{uh(u)}. \tag{59}$$

Thus $(r^2-1)\cos 2\varphi$ is invisible to this statistic at first order for every $u > 0$. The potentials differ, and their images, fluxes and delays generally differ. The cancellation occurs only after compressing those data into $M_{\rm app}$. In particular, a leading screened $\epsilon\cos 2\varphi$ produces the same apparent-mass history as a conventional shear $\gamma = 2\epsilon$ of the same orientation. Different radial phase powers alone do not establish identification.

For the radially constant phase, the two primary-image angles differ by π, so $p(x_+) = p(x_-)$ and $\delta\Delta\tau = 0$. Nevertheless $\delta \ln R = -4\epsilon s\cos 2\varphi_s/u$, and tangential astrometry can also change. At $\cos 2\varphi_s = 0$ even the flux/mass statistic is blind at first order while tangential shifts remain possible. Full transition matching restores a small radial and delay response (Figure 2); this does not make the leading degeneracy irrelevant.

6 Joint primary-pair information experiment

6.1 Declared data and nuisance family

The numerical experiment uses the action-derived scalar monopole as well as the quadrupole, rather than fitting a point-mass background to an unlabeled modified profile. Its observables at

each of nine reference configurations are

$$O_i = (x_{+,i}, y_{+,i}, x_{-,i}, y_{-,i}, \ln R_i, \Delta t_i/t_M). \tag{60}$$

The source reference positions are $(1.8t_i, 0.6)$, with t_i uniformly spanning $[-1, 1]$. In the fit, two source coordinates are *free at every configuration*; no known unlensed track is supplied. They can absorb parallax, source motion and source-position errors at these sample points. The lens nuisances are one mass, two center coordinates, two constant-velocity coordinates, one convergence, two external-shear and two compact material-quadrupole coordinates. The scalar flux is also fitted through its local monopole response, rather than assumed known in the inference. The visible axial scalar quadrupole is also fitted as a radial nuisance with its normalized Y_{20} response. There are 30 nuisance parameters and two transverse envelope coordinates in 54 data components. The two odd projected tensor modes are absent from this one-view data model rather than artificially fixed by a prior.

The ordinary quadrupoles are $B_A r^{-2}$ and $\Gamma_A r^2$ in the cosine/sine basis. They are fixed as a family before inspecting the envelope projection. Internal and external radial multipole information is familiar in lens modeling [11]; it is not supplied here as a new general principle. The inference assumes known redshifts, distances, time calibration and a common absolute astrometric frame. Unknown frame drift is equivalent to part of the freely fitted center motion; arbitrary frame distortion or an unrestricted radial potential is not covered. Fluxes are intrinsic geometric point-source fluxes with no extinction, blending, independent microlensing or finite-source structure. These are strong experimental assumptions, not observational facts.

For each baseline image, $A = I - D^2\psi_0$. The computed response includes

$$\delta x = A^{-1}(\nabla\delta\psi + \delta y), \qquad \delta \ln|\mu| = \mathrm{tr}\Big[A^{-1}(D^2\delta\psi + D(D^2\psi_0)[\delta x])\Big]. \tag{61}$$

For a translated center, $\delta\psi = -\delta c \cdot \nabla\psi_0$. Mass differentiation changes the action's scalar monopole too. Source variation enters the delay as $(x_+ - x_-) \cdot \delta y$, in addition to the potential difference. These terms ensure comparison of consistent physical states and prevent spurious source/center signatures.

Let N and S denote the nuisance and envelope derivative matrices. With a specified data covariance C_O, local information is the Gram matrix of

$$S_\perp = (I - P_N)C_O^{-1/2}S, \tag{62}$$

where P_N projects onto the column space of $C_O^{-1/2}N$. This is ordinary profiled linear inference. Here all data components have equal *dimensionless* errors in the displayed units, an explicit choice for information comparison. It is not an instrument noise model. Nuisance columns are normalized before SVD; the recorded rank threshold is 10^{-9} of the largest singular value, and the full spectra are released. Raw column norms below 10^{-11} times the larger of the maximum norm and unity are discarded as numerical zeros. All 30 full-pair columns survive that admission. The axial mode is nearly masslike; when all admitted columns have full rank but condition number exceeds 10^6, the projection uses 70-digit normal equations after the unchanged SVD rank selection. An independent 70-digit QR solve checks the result. These refine arithmetic for the same double-input derivatives, not the physical approximation. Parameter priors are not added to fill nulls.

6.2 Evaluated response and controls

For $M = 10^9 M_\odot$, $z_l = 0.3$, $z_s = 1$, $r_0 = 10$ pc and scalar flux proxy equal to the regular-body value, the same-background distances give $r_E = 339.62$ pc, $\theta_E = 74.116$ mas and $t_M = 25612.55$ s. The computed screening radius is $r_V = 45680.31$ pc. This scale is action-derived after choosing the matching flux; it is not tuned to the rays.

Table 2: Declared conditional example and local information result. The last two rows use the small supplied state at source $(0.9, 0.6)$; the singular values use unit optical phase amplitudes and the stated dimensionless data metric.

Quantity	Value
Mass	$10^9 M_\odot$
Worldtube / Einstein radius	10.0 / 339.62 pc
Screening radius	45.68031 kpc
Worldtube compactness	4.79×10^{-6}
Boundary amplitude / optical amplitude	1.38×10^{-16} / -8.89×10^{-13}
Joint-pair nuisance rank	30
Projected envelope singular values	3.13473, 0.80220
Retained column-norm fractions	0.04213, 0.14512
Largest tested physical coordinate shift	9.36×10^{-5} nanoarcsec
Tested physical log-flux-ratio shift	2.88×10^{-12}

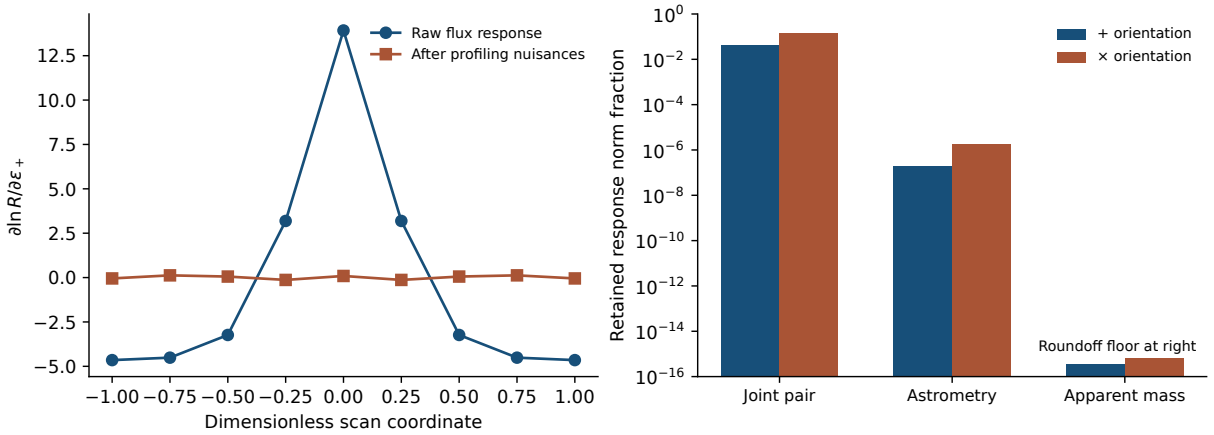


Figure 3: Information retained by the declared broad scan. Left: the flux component of the + response before and after simultaneous nuisance projection in the full data vector. Right: retained column-norm fractions for three data choices; the apparent-mass residual is at roundoff. Comparisons use the stated dimensionless metric and nuisance family, not a telescope covariance or a universal observability theorem.

The joint-pair projected envelope singular values are approximately $(3.135, 0.802)$ per unit phase amplitude. Roughly 4–15% of each envelope column's norm survives the 30 nuisance directions. The strict constant-phase replacement gives nearly the same joint-pair result; identification here therefore comes mainly from keeping the separate image/flux/delay information rather than resolving the small transition correction. Astrometry alone retains only very small directions in this example, and its near-null spectrum is numerically delicate. The apparent-mass statistic alone retains no measurable independent direction with the admitted free-source/action-monopole family. The analytic point-mass/shear null in Equation (59) separately establishes an exact first-order loss without relying on an SVD cutoff.

Controls vary mass, worldtube radius and the supplied scalar flux. With zero spherical scalar flux the exterior quadrupole is harmonic, $f_2 \propto r^{-3}$; its transverse projected phase is exactly an ordinary compact quadrupole and the retained response vanishes to numerical precision. Nonzero fluxes 0.1, 1 and 3 times the proxy give a screened response with the same broad-scan information pattern, as do the tested 10^6 and $10^{11} M_\odot$ examples with declared weak-field worldtubes. These are conditional boundary families, not alternative theories or inferred black-hole charges. Allowing an independent center at every configuration reduces the joint information but does not remove both transverse directions in this restricted family. Arbitrary time-dependent quadrupoles would

be a different, less constrained problem.

6.3 Perturbative and numerical control

The physical state example is chosen by an explicit small parameter,

$$a_+ = \frac{\eta x_0 |p(x_0)|}{6}, \quad a_\times = 0, \qquad \eta = 10^{-3}, \quad x_0 = r_0/L_0. \tag{63}$$

The factor six is the quadrupole angular eigenvalue; the numerical checks also evaluate actual Hessian ratios and the quadratic residual in the retained nonlinear scalar equation. This is a supplied perturbatively small state, not a BHSM-selected amplitude or an upper bound on astrophysical deformations. It gives $a_+ \simeq 1.38 \times 10^{-16}$ and $\epsilon_+ \simeq -8.89 \times 10^{-13}$.

At source $(0.9, 0.6)$ the retained model gives a largest coordinate shift of about 9.4×10^{-5} nanoarcsec and $\delta \ln R \simeq 2.88 \times 10^{-12}$. The formal matched delay shift is approximately 5.1×10^{-12} s. That last small number is a value in the truncated static thin-lens model, whose leading delay term vanishes; it is *not* a physically certified picosecond forecast. Uncomputed time-dependent and matching corrections require their own comparison before interpreting such a subleading delay.

The code checks the radial equation against the special-function solution, the dipole translation mode, independent original-z versus transformed line integrals, finite-difference projected derivatives, boundary energy normalization, and finite nonlinear primary-image continuations. At test amplitudes decreasing from 10^{-3} to 3×10^{-5}, the central-difference error in the full response decreases quadratically to about 1.6×10^{-7}, with image stationarity residuals below 10^{-14}. Those larger optical test amplitudes test the lens derivative; they do not validate the small-state scalar linearization at an enlarged boundary amplitude. Independent symbolic/Cartesian and polynomial-branch checks verify the estimator operator and its nulls. All are internal AI-assisted checks.

Quintic interpolation uses analytic projected derivatives at logarithmic nodes; measured midpoint discrepancies and quadrature refinements are recorded as numerical diagnostics, not interval certificates. The formal infinite exterior is matched to the frozen local background. At a 1 Mpc cutoff, explicit far-tail bounds give a monopole deflection tail below 9×10^{-11} of the ordinary deflection and a transverse quadrupole phase tail below 8×10^{-11} pc per unit boundary amplitude. These bound omitted tails only. They do not bound every weak-field, cosmological or effective-theory correction.

7 The runnable Blackhole Tracker product

The released research product makes the compact-lens measurement task operational. A local browser app, command line, importable Python package and two executed notebooks share the same versioned data contract and inference engine. This is an offline analysis tool for supplied resolved primary-pair measurements, not a sky-survey acquisition service. Its ordinary-lens mode can estimate a candidate's mass, center and motion without requiring a BHSM deviation. Its optional R1 mode evaluates the conditional exterior and the admission bound of Section 3. Established lens software already supports sophisticated lens inference: `lensmodel`, `lenstronomy` and `PyAutoLens` [17–19]; the moving-lens `BAGLE` framework is a particularly close predecessor [20]. No first mass/motion tracker or general replacement for those tools is claimed.

7.1 Measurement contract and likelihood

An input supplies angular-diameter distances D_l, D_s, D_{ls}, redshifts, a reference observer epoch, and N epoch/source rows, with $3 \le N \le 80$. Each row contains two resolved image positions in

mas, $\ln(|\mu_+|/|\mu_-|)$ and the negative-parity minus positive-parity arrival-time difference in seconds. The labels refer to parity, not a brightness ordering; a perturbed flux ratio may be below unity. The covariance is the full positive-definite $6N \times 6N$ matrix in row-major observable order. Neither time sorting nor diagonalization is applied. Correlated astrometric, flux and timing errors therefore remain coupled under Cholesky whitening. A provenance field explicitly declares synthetic or observed measurements. Validation of that field and the numerical schema does not verify an observation's origin.

With supplied parallax factors $\boldsymbol{P}_i$, the lens center is

$$\boldsymbol{c}_i = \boldsymbol{c}_0 + \boldsymbol{\mu}_{\rm obs}(t_i - t_0) + \frac{1000}{D_l/\mathrm{pc}}\boldsymbol{P}_i \ \mathrm{mas}. \tag{64}$$

Omitting these factors means a declared zero observer-parallax term. Distances, the astrometric frame and timing calibration are treated as known; they are not silently inferred from the lens data. The source position has two free coordinates at every row, even when several rows represent repeated observations of one source. Source trajectory truth is never supplied to the fitter. The ordinary model contains a point mass and constant center motion, with selectable shear or the larger family of shear, convergence and compact quadrupole. It solves the nonlinear lens equation for the continued positive/negative primary branches, rejects critical or incorrectly ordered branches, and uses their full magnification determinants and Fermat delay.

The optimizer minimizes

$$\chi^2(\vartheta) = \|L^{-1}[d - f(\vartheta)]\|_2^2, \qquad LL^T = C. \tag{65}$$

An analytic point-lens delay/flux initializer supplies a computational mass scale, followed by deterministic perturbed starts and bounded nonlinear least squares. The default mass search domain is $[10^{-3}, 10^{12}]M_\odot$, exposed in the input rather than confused with a detection prior. A conservative slow-motion domain is declared separately. Final solutions undergo branch, worldtube, weak-field and motion checks. Local covariance follows the whitened Jacobian SVD, without rescaling the supplied C by reduced χ^2. Null directions have unavailable uncertainties. Center-track bands propagate the center–motion cross covariance. Reoptimized one-coordinate likelihood profiles are available from the command line and notebook; their usual $\Delta\chi^2$ reference values do not guarantee nonregular coverage.

7.2 Product decisions and the conditional extension

`FIT_REJECTED` denotes solver failure, unacceptable residuals at the declared $p < 0.001$ guard, or failure of a physical-domain check. `INSUFFICIENT_INFORMATION` denotes unavailable or very weak mass information. Otherwise, `PARTIALLY_IDENTIFIED` reports weak motion or a global parameter at its search bound. `TRACKED` requires a supported mass and a two-dimensional local motion Mahalanobis radius of at least five, together with the other checks. This is a transparent product decision rule within the supplied model and covariance; it is not a black-hole classification or a universal false-alarm probability. Every result retains its full residuals, assumptions and diagnostics.

In R1 mode the spherical flux is either the explicitly labeled fixed regular-body proxy or a fitted conditional matching datum. The latter is bounded to the declared fitted domain $0.05 \le \chi \le 20$; discrete runtime comparisons do not certify that entire interval. Neither option is called horizon matched. After fitting the monopole and ordinary nuisances, unit-amplitude transverse responses are whitened and projected off the fitted Jacobian and the visible axial response. If their singular values are $s_1 \ge s_2$, the diagnostic compares $\mathcal{A}_{\max} s_j$ with unity. A rank loss or best admitted response below unity returns `UNRESOLVED`. The alternative label, `INFORMATION_AVAILABLE`, denotes local conditional sensitivity, not a detected deformation. A formal unconstrained state outside

the cap is explicitly marked outside the admitted family. The product does not inflate it into a physical fit.

For the noisy nearby fixture below, the ordinary fit gives $M = 99947.2 \pm 38.7\,M_\odot$ and $\boldsymbol{\mu} = (0.06667, -0.03808)$ mas/yr with local component errors $(0.00119, 0.00123)$ mas/yr; $\chi^2 = 36.39$ for 29 degrees of freedom. These are one synthetic realization's outputs. For R1 analysis with the fixed regular-body flux proxy and shear nuisances, that fixture can retain the ordinary track while returning an unresolved envelope: the admitted state cap is about 5.34×10^{-21} and the largest projected admitted local response-to-noise is about 2.33×10^{-14}. Thus making the tracker usable does not supply evidence for the proposed BHSM deformation signal.

The app supports input inspection, validation, fitting, track and observable plots, uncertainty/null diagnostics and JSON/HTML export. Command-line export additionally supplies prediction CSV, a static track/residual figure and SHA-256 manifest. The first notebook reproduces mass/motion inference, a mass profile and the weak-motion control; the second evaluates conditional response and admission. Raw theory inputs are packaged and hash checked, so these interfaces do not depend on a private archive path or the author's checkout. A separate stellar-mass example adapts rounded parameters from Lu et al. [20] into a declared synthetic primary-pair test with a chosen observer ellipse. Its positions, fluxes and added delays are generated by this product. No `BAGLE` engine was run, and no observed data or real ephemeris was validated. The independent ordinary-lens checks instead evaluate the analytic point-mass equations in physical units separately from the package's lens helpers. The repeated-noise experiment tests recovery and local interval coverage under the product's own declared synthetic model.

8 Nonlinear product validation with known covariance

The software product is tested separately from the preceding 54-observable theory experiment. Its ordinary compact-lens inference fits mass, reference center and observer proper motion, two shear components, and two free source coordinates at every row. Distances, redshifts and the absolute astrometric frame are supplied calibrations. The baseline validation is entirely synthetic; its adopted errors do not specify an instrument capability. Optional R1 information retains the separate conditional-exterior and physical-amplitude restrictions. The frozen nearby fixture contains a $10^5 M_\odot$ lens at $D_l = 1$ Mpc, three background source positions sampled at each of three epochs over ten observer years, and true motion $(0.065, -0.038)$ mas/yr. The nine rows supply 54 image-coordinate, log-flux-ratio and delay measurements. Their declared base noise scales are 0.02 mas, 0.005 and 3×10^{-4} s, respectively, with supplied astrometric and timing correlations. The full covariance C is retained in $\chi^2 = (d-f)^T C^{-1} (d-f)$; errors are not rescaled by the fitted reduced χ^2. These inputs test a resolved primary-pair model with free source positions, rather than assume a known unlensed source track.

The noiseless recovery control (C6) uses the data-derived initializer and two fixed perturbations, including mass factors 1.5 and 0.65 and shifts of center, motion, sources and shear. Each fit uses three internal starts. All recover the mass, center and motion within the frozen relative or reference-scale tolerances 10^{-5}; the largest χ^2 is 3.01×10^{-23}, below the fixed 10^{-6} criterion. Truth is used to score recovery, not supplied as an initializer. This checks nonlinear solver recovery for this fixture, not global likelihood uniqueness.

8.1 A fixed repeated-noise calibration

Control C7 draws 64 independent Gaussian noise vectors using the full C, a single frozen PCG64 stream seeded 4932026, and one start per ordinary-lens fit with shear admitted. Every realization remains in the record and coverage denominator; an exception, rejected fit or unavailable interval would count as a miss. All 64 fits converge and return the product's `TRACKED` status, with no

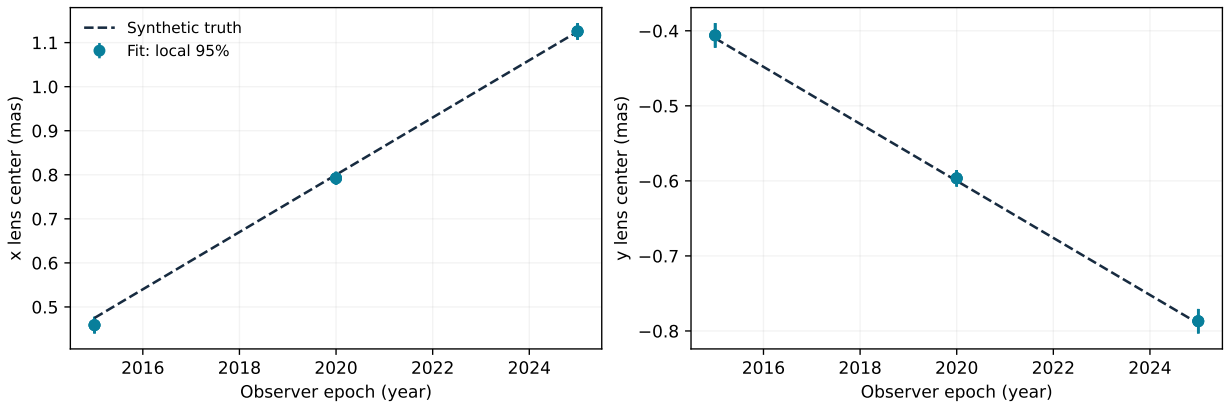


Figure 4: The noisy nearby synthetic fixture: recovered lens-center coordinates with local 95% marginal intervals propagated from the full center–motion covariance, against truth used only for scoring. Repeated rows at each epoch overlap. This is a ten-year three-source product test, distinct from the broad single-track information scan.

Table 3: Frozen 64-draw local 95% interval coverage. Each exact two-sided binomial test has null coverage 0.95 and rejection level $\alpha = 0.01$. All five tests are reported without a multiple-testing adjustment.

Coordinate	Covered / 64	Fraction	Exact p
Mass M	61	0.9531	1.0000
Center x	59	0.9219	0.2532
Center y	61	0.9531	1.0000
Motion x	61	0.9531	1.0000
Motion y	61	0.9531	1.0000

excluded failures. The reported intervals are local known-covariance Gaussian intervals, not reoptimized likelihood profiles. For mass we use $\widehat{M}\exp[\pm 1.959964\,\sigma_M/\widehat{M}]$; center and motion use $\widehat{\theta} \pm 1.959964\,\sigma_\theta$.

None fails the preregistered gross-miscalibration guard (Table 3). The exact 95% binomial interval for 61/64 coverage is $[0.8691, 0.9902]$, and for 59/64 it is $[0.8270, 0.9741]$: 64 draws provide a limited calibration check. The receipt also retains parameter bias and scatter. In particular, the mean y-motion error is 3.98×10^{-4} mas/yr, about 2.75 estimated standard errors of its ensemble mean. Passing the frozen coverage test does not establish negligible bias or universal coverage.

8.2 Weak information, model mismatch and finite cause ambiguity

The negative controls preserve unsuccessful scientific outcomes (Table 4). The ten-year cosmological fixture (C8) has true motion $(5, -3) \times 10^{-5}$ mas/yr and returns `PARTIALLY_IDENTIFIED`, with motion signal-to-noise 0.00482 and component errors $(0.0121, 0.0124)$ mas/yr. Its mass information does not license a resolved motion claim. For a separate conventional-shear fixture (C11), omitting shear gives a rejected fit at the frozen goodness-of-fit criterion $p < 0.001$; admitting shear recovers the declared $(0.03, -0.018)$ coefficients. Residual structure is therefore not assigned automatically to an R1 envelope or a new physical cause.

For C9 the zero-flux harmonic boundary response is exactly an ordinary compact r^{-2} quadrupole, with coefficient $-\widehat{\beta} N_2 r_0^3/[3m_{\mathrm{ref}}(r_E^{\mathrm{ref}})^2]$ per boundary amplitude, in the declared reference lens units. Three finite operator comparisons and compensating coefficient changes agree in potential, gradient and Hessian to 2.22×10^{-16}. Four combined compact/boundary columns have rank two. Large formal coefficients are confined to this operator identity test; public prediction separately

Table 4: Controls with scientifically meaningful weak or null outcomes.

Control	Quantity	Outcome
C8: short cosmological track	Motion SNR 0.00482	Unresolved motion
C11: shear omitted	$\chi^2 = 28297.8$	Fit rejected
C11: shear admitted	$\chi^2 = 1.95 \times 10^{-22}$	Baseline recovered
C9: harmonic / compact pair	Four columns, rank two	Two exact nulls

rejects that nonadmitted state. They are not enlarged physical signals. The current nonlinear fit does not expose a joint free fit of both degenerate amplitude pairs, so separate-amplitude interval calibration is not claimed. These controls accompany the released independent interface, physical-unit, covariance and admission tests. Versioned JSON receipts retain every draw, initializer, criterion, runtime and SHA-256 source hash; the R1 provider inputs are pinned and checked on loading. The scripts `calibrate_product.py` and `check_finite_harmonic_degeneracy.py` reproduce the controls. The validated scope is the declared synthetic compact-lens families and local intervals. Unknown calibration, broader environmental freedom, critical image configurations, observational sensitivity and horizon identification remain outside this validation.

9 What could actually be tracked?

The product demonstrates compact-lens inference under its declared synthetic measurement and covariance model. It does not demonstrate observational acquisition or a detectable BHSM envelope. The earlier broad information scan has a distinct limitation. For the illustrative cosmological mass and transverse speed 300 km s^{-1}, ten observer years move the center only 0.00236 pc, or $6.95 \times 10^{-6} r_E$. Traversing the broad synthetic $3.6 r_E$ scan at that speed would take about 5.2 million observer years. The short-span design has nearly dependent columns; its released spectrum must not be promoted into a realistic parameter forecast. Several independent source configurations could sample a broad spatial range, but that would require an actual source population, their redshifts and a corresponding data model. It is not automatically supplied by nine synthetic points.

A translated static profile is an adiabatic template evaluated at lens crossing time. It requires slow motion, control of coefficient evolution over the relevant crossing times, and control of scalar propagation. For the speed example $v/c \simeq 10^{-3}$, even nominal relative corrections at that level cannot be ignored when claiming the tiny subleading delay. Positive leading principal coefficients establish the local branch's hyperbolicity/ellipticity conditions; they do not prove global nonlinear stability, a UV completion or horizon admissibility. Fast intrinsic deformation or ringdown needs a causal metric solution. Such time-dependent black-hole lensing already has separate explicit models [21]; they do not close the R1 matching problem.

The unknown flux is especially consequential. The regular-body integrated source equation supplies one proxy, whereas a black-hole exterior has an integration constant that must be selected by horizon, interior and cosmological boundary conditions. The searched BHSM record did not establish that selection. A surface-acceleration identity or a Brown–York energy inversion on a known Schwarzschild worldtube does not determine this scalar flux or its deformation response. Nor can the two odd line-of-sight tensor modes be reconstructed by improving the precision of the same static projection.

This missing evaluation is specific to the present action and background. Cosmologically induced black-hole charges have been derived in related shift-symmetric Horndeski models [22], and regular stable quasi-stationary cubic-Galileon black holes with cosmological asymptotes have been constructed [23]. Those works provide a concrete matching strategy. The explicit self-accelerating cubic de Sitter constructions are not the present canonical-sign, linear-potential,

rolling matter background; directly substituting the latter coefficients into that self-accelerating branch does not give a real Hubble parameter. Their results therefore cannot simply select the flux proxy used here, nor do they provide this paper's normalized quadrupole state.

Even perfect matching and a detectable transfer would track an exterior compact-mass/envelope combination. Any alternative interior producing the same Weyl response on every sampled ray gives the same data. A horizon claim therefore needs additional physics or independent observables. Unrestricted source transformations, line-of-sight potentials, uncertain distances, evolving macrostructure and flux propagation effects can also change the finite-family inference result. Multiple source redshifts alone do not remove every such freedom.

10 Scientific advance and limits

Static cubic screening, its multipole powers and exact Green functions are established [5–8]. General STF gravitational multipole projection is also established [9]. The point-mass delay/flux estimator is standard, and ordinary radial multipole separation and profiled linear inference are standard methods. We do not claim priority for those ingredients.

The addition combines their normalized R1 optical composition with an exterior-wide sufficient state cap and a runnable compact-lens inference workflow. Its theory exposes a delay-blind leading response, an exact apparent-mass/shear degeneracy and line-of-sight nulls. Its product retains full covariance and source freedom, and reports calibrated synthetic recovery together with weak-motion, mismatch and physically unresolved outcomes. The finite harmonic control makes a cause ambiguity exact rather than assigning every structured residual to BHSM.

The result supports a theoretical and computational lens-modeling methods paper with a usable specialist research tool. It does not yet establish a detectable BHSM deformation, horizon classification or a decisive new astrophysical discovery. Further R1 prediction needs a normalized black-hole/homogeneous-background boundary map selecting exterior flux and deformation dynamics. Observational use additionally requires actual resolved pairs, calibrated geometric fluxes and delays, and a model adequate for their environment. No telescope detection or journal submission is reported.

Data and code availability. The complete source package contains the three preserved research projects, this manuscript, the `product/` package, executed notebooks, synthetic examples, source pins and numerical/review receipts. The paper's theoretical replay entry point is `scripts/reproduce.py`; the product's is `product/scripts/validate_product.py`. Its README documents app, command-line and Python use and the data contract. All validation inputs are synthetic or theoretical; no fresh observational validation set is used. The released source and draft revision are available in the author's repository draft.

Author and AI disclosure. The manuscript was prepared for Norman P. Carberry with AI assistance for recovery, algebra, implementation, literature checking and drafting. Independent agent checks are internal computational reviews, not external peer review. Scientific authorship and submission require human review. No external funding or conflict is asserted by the calculation.

A First variation of the delay–flux mass estimator

The point-mass delay relation is standard; see, for example, Muñoz et al. [12]. This appendix derives its first variation under a general local lens-potential perturbation, including the displacement of the images in the magnification calculation. The result is a thin-lens projection rule; it selects neither a horizon response nor a physical surface-state amplitude.

Use the reference point-mass Einstein angle and observed time scale

$$\theta_E^2 = \frac{4GM}{c^2}\frac{D_{ls}}{D_l D_s}, \qquad t_M = \frac{4GM}{c^3}(1+z_l),$$
$$\boldsymbol{x} = \boldsymbol{\theta}/\theta_E, \quad \boldsymbol{y} = \boldsymbol{\beta}/\theta_E. \tag{66}$$

Hold the physical source, body mass, distances, redshift and screen fixed while varying the dimensionless potential,

$$\tau_\epsilon(\boldsymbol{x};\boldsymbol{y}) = \tfrac{1}{2}|\boldsymbol{x}-\boldsymbol{y}|^2 - \ln r - \epsilon p(\boldsymbol{x}), \qquad r = |\boldsymbol{x}|. \tag{67}$$

For $u = |\boldsymbol{y}| > 0$, put $s = \sqrt{u^2+4}$. On the source axis the reference images are $x_+ = (u+s)/2 > 1$ and $x_- = -1/x_+ < 0$. Continue the positive-parity minimum and negative-parity saddle without crossing a critical curve or reversing their flux/time ordering. With $R = |\mu_+|/|\mu_-| > 1$ and $\Delta t = t_M(\tau_- - \tau_+) > 0$,

$$\ln R_0 = 4\,\mathrm{asinh}(u/2), \qquad h(u) = \tfrac{1}{2}us + 2\,\mathrm{asinh}(u/2),$$
$$M_{\mathrm{app}} = \frac{c^3\Delta t}{4G(1+z_l)h[2\sinh(\frac{1}{4}\ln R)]}. \tag{68}$$

Indeed, the lens equation $x - u - 1/x = 0$ gives $\mu_+ = x_+^4/(x_+^4-1)$, $\mu_- = -1/(x_+^4-1)$ and $\tau_- - \tau_+ = \frac{1}{2}(x_+^2 - x_+^{-2}) + 2\ln x_+ = h(u)$. The distance factors in the physical time prefactor cancel against θ_E^2, so $M_{\mathrm{app}} = M$ for every reference alignment. The aligned Einstein-ring limit is excluded from the two-image estimator.

Let $\delta = \partial_\epsilon|_0$ and define

$$H_0(\boldsymbol{x}) = D^2 \ln r = \frac{I}{r^2} - \frac{2\boldsymbol{x}\boldsymbol{x}^T}{r^4}, \qquad A_i = I - H_0(\boldsymbol{x}_i). \tag{69}$$

Differentiating the stationary lens equation at fixed $\boldsymbol{y}$ yields

$$\delta\boldsymbol{x}_i = A_i^{-1}\nabla p(\boldsymbol{x}_i), \qquad \delta A_i = -D^2 p(\boldsymbol{x}_i) - DH_0(\boldsymbol{x}_i)[\delta\boldsymbol{x}_i],$$
$$\delta \ln|\mu_i| = -\operatorname{tr}(A_i^{-1}\delta A_i). \tag{70}$$

The evaluation-point term is essential. Explicitly,

$$DH_0(\boldsymbol{x})[\boldsymbol{v}] = -\frac{2(\boldsymbol{x}\cdot\boldsymbol{v})I}{r^4} - \frac{2(\boldsymbol{v}\boldsymbol{x}^T + \boldsymbol{x}\boldsymbol{v}^T)}{r^4}$$
$$+ \frac{8(\boldsymbol{x}\cdot\boldsymbol{v})\boldsymbol{x}\boldsymbol{x}^T}{r^6}. \tag{71}$$

Stationarity removes image displacements from the first arrival-time variation, giving $\delta(\tau_- - \tau_+) = p(\boldsymbol{x}_+) - p(\boldsymbol{x}_-)$. Since $h' = s$ and $(\ln R_0)' = 4/s$, the total estimator variation is

$$\boxed{\delta \ln M_{\mathrm{app}} = \frac{p(\boldsymbol{x}_+) - p(\boldsymbol{x}_-)}{h(u)} - \frac{s^2}{4h(u)}[\delta\ln|\mu_+| - \delta\ln|\mu_-|].} \tag{72}$$

Thus a pure alignment displacement cancels. A changing physical mass or redshift prefactor is a separate variation; it must not be hidden in an unspecified potential normalization.

For the quadrupole $p = q(r)\cos[2(\varphi - \varphi_q)]$, the two reference image angles differ by π. Write $c_2 = \cos[2(\varphi_s - \varphi_q)]$, $r_+ = (u+s)/2$, $r_- = 1/r_+$, $a_r = 1 + r^{-2}$ and $a_t = 1 - r^{-2}$. The radial image displacement is $\delta x_r = q' c_2/a_r$. Equation (70) then gives

$$\delta\ln|\mu_i| = c_2\mathcal{F}_q(r_i), \tag{73}$$

$$\mathcal{F}_q(r) = \frac{q''}{a_r} + \frac{q'/r - 4q/r^2}{a_t} + \frac{2q'}{r^3 a_r}\left(\frac{1}{a_r} - \frac{1}{a_t}\right), \tag{74}$$

$$\delta\ln M_{\mathrm{app}} = c_2\mathcal{K}_q(u), \tag{75}$$

$$\mathcal{K}_q(u) = \frac{q(r_+) - q(r_-)}{h(u)} - \frac{s^2}{4h(u)}[\mathcal{F}_q(r_+) - \mathcal{F}_q(r_-)]. \tag{76}$$

The final term of $\mathcal{F}_q$ is the radial evaluation-point contribution. Tangential displacements contribute off-diagonal terms whose trace vanishes in the reference radial/tangential eigenbasis. The physical action or worldtube source must supply q, q', q'' in the same lens units before this linear projection is applied.

The following closed forms provide nuisance comparisons and independent checks. Defining $K_* = s^3/[uh(u)]$, direct substitution yields

$$\mathcal{K}_{r^2} = K_*, \qquad \mathcal{K}_{r^{-2}} = (u^2+1)K_*, \qquad \mathcal{K}_1 = K_*. \tag{77}$$

Hence a compact projected quadrupole $Br^{-2}\cos[2(\varphi-\varphi_B)]$ contributes $B(u^2+1)K_*\cos[2(\varphi_s-\varphi_B)]$, while an external shear in the convention $(\gamma/2)r^2\cos[2(\varphi-\varphi_\gamma)]$ contributes $(\gamma/2)K_*\cos[2(\varphi_s-\varphi_\gamma)]$. At one alignment the amplitudes are degenerate; several alignments distinguish these two specified shapes only to the extent that the nuisance assumptions hold.

Linearity also gives a nontrivial blind direction,

$$\boxed{\mathcal{K}_{r^2-1}(u) = 0 \qquad \text{for every } u > 0.} \tag{78}$$

This is an exact *first-order* null of the statistic, not an exact finite-perturbation equivalence or a vanishing lens deformation. The potential $(r^2-1)\cos 2\varphi$ generally changes images, delay and flux ratio. It is a local smooth-potential counterexample on image annuli; whether a specified physical source can generate it requires its own source/boundary analysis. Different radial potentials therefore do not automatically give identifiable tracker signals. An inference must compare the evaluated $\mathcal{K}_q$ with the justified nuisance response space, including its conditioning, rather than compare potential labels.

These formulas concern quasistatic geometric optics and two noncritical continued primary images. They do not describe near-horizon strong-deflection images, new-image total flux, differential extinction or unmatched variable source emission. Exact symbolic identities, Cartesian total-Hessian checks and finite on-axis polynomial branch checks are supplied with the reproduction sources; they are internal consistency tests, not observations.

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